

Abhinav Shrestha

Chair of Ecosystem Dynamics and Forest Management in Mountain Landscapes
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EDUCATION

University of Idaho <i>Master of Science in Geography</i>	Moscow, ID, USA 2021 – 2023
Coe College <i>Bachelor of Arts in Biology & Environment Science</i> <i>Magna Cum Laude, Phi Beta Kappa, Phi Kappa Phi</i>	Cedar Rapids, IA, USA 2015 – 2019

PUBLICATIONS & DATA SETS

- Shrestha, A.**, Hicke, J.A., Meddens, A.J.H., Karl, J.W., Stahl, A.T., (2024). Evaluating a Novel Approach to Detect the Vertical Structure of Insect Damage in Trees Using Multispectral and Three-Dimensional Data from Drone Imagery in the Northern Rocky Mountains, USA. *Remote Sensing*, 16, 1365. <https://doi.org/10.3390/rs16081365>
- Harrison, G. R., **Shrestha, A.**, Strand, E.K., Karl, J.W., (2024). A comparison and development of methods for estimating shrub volume using drone-imagery-derived point clouds. *Ecosphere* 15, e4877. <https://doi.org/10.1002/ecs2.4877>
- Hicke, J. A., Bright, B. C., Hanavan, R. P., Hudak, A. T., Meddens, A. J., **Shrestha, A.**, & Stahl, A. T. (2024). Remote sensing of forest insect and disease outbreaks in the western United States: Tree, stand, and landscape responses and technologies and methods for detection and attribution. *RMRS-GTR-443*. Fort Collins, CO: US Department of Agriculture, Forest Service, Rocky Mountain Research Station. 92 p., 443. <https://doi.org/10.2737/RMRS-GTR-443>
- Zimmer, S. N., Houtman, R. M., Leatherman, L. S., Housman, I. W., **Shrestha, A.**, Shaw, J. D., & Riley, K. L. (2025). *TreeMap 2020 CONUS: A tree-level model of the forests of the conterminous United States circa 2020*. (Fort Collins, CO: Forest Service Research Data Archive)[Data set]. <https://doi.org/10.2737/RDS-2025-0031>
- Houtman, R. M., Leatherman, L. S., Zimmer, S. N., Housman, I. W., **Shrestha, A.**, Shaw, J. D., & Riley, K. L. (2025). *TreeMap 2022 CONUS: A tree-level model of the forests of the conterminous United States circa 2022*. (Fort Collins, CO: Forest Service Research Data Archive)[Data set]. <https://doi.org/10.2737/RDS-2025-0032>

THESES

- Shrestha, A.** (2023). *Combining multispectral and three-dimensional data from drone imagery to detect forest insect damage: an evaluation of a novel approach to identify the vertical structure of damage in trees in the northern Rocky Mountains, USA* [Masters thesis, University of Idaho]. University of Idaho Library. <https://verso.uidaho.edu/esploro/outputs/996638033701851>
- Shrestha, A.** (2019). *GIS-based study of topographical preference of common tree species in Palisades-Kepler State Park, IA* [Bachelors thesis, Coe College]. Coe College Stewart Memorial Library. <https://coecollege.on.worldcat.org/oclc/1137317800>

SELECTED PRESENTATIONS

- Riley, K., Houtman, R., Zimmer, S., Leatherman, L., Housman, I., **Shrestha, A.** (2023). *TreeMap and FuelMap: imputed tree and downed woody material information from the FIA databases for the forests of CONUS, background and methodological updates in the 2020 and 2022 datasets* [Conference presentation]. USDA Forest Service Forest Inventory and Analysis (FIA) Science Symposium.
- Shrestha, A.**, Hicke J. A., Meddens A. J. H, Karl J. W., Stahl A. T. (2024). *Evaluation of a novel approach to detect the vertical structure of insect damage in trees using multispectral and three-dimensional data from drone imagery in the northern Rocky Mountains, USA* [Poster presentation]. 72nd Western Forest Insect Work Conference, Missoula, MT.
- Harrison, G. R., **Shrestha, A.**, Karl J. W. (2023). *Seeing shrubs from the sky: an exploration of using drone-based method to estimate shrub canopy volume* [Conference presentation]. Ecological Society of America (ESA) Annual Conference, Portland, OR.
- Hicke, J. A., **Shrestha, A.**, A. Meddens, A. Stahl, B. Bright, A. Hudak, R. Hanavan (2023). *Using Remote Sensing to Study Tree Damage from Insects* [Conference presentation]. 71st Western Forest Insect Work Conference, Seattle, WA.
- Shrestha, A.**, and Sanchini P (2019), *A Geographic Information System (GIS)-based study on topographical preference of common tree species in Palisades-Kepler State Park* [Conference presentation]. Coe College Student Research Symposium, Coe College, Cedar Rapids, IA.
- Obiesie, C., **Shrestha, A.**, & Sanchini, P (2018). *Distribution of Oaks (Quercus spp.) in Forests of Palisades-Kepler Park, Linn County, Iowa* [Poster presentation]. Coe College Student Research Symposium, Coe College, Cedar Rapids, IA.

SCHOLARSHIPS, AWARDS, & HONORS

Outstanding Graduate Student – Teaching	2023 – 2024
<i>University of Idaho Dept. of Earth & Spatial Sciences, Moscow, ID, USA</i> Award for graduate teaching assistants.	
Outstanding Graduate Student – Research	2022 – 2023
<i>University of Idaho Dept. of Earth & Spatial Sciences, Moscow, ID, USA</i> Award for graduate students conducting research.	
Graduate Assistantship	2021 – 2023
<i>University of Idaho Dept. of Earth & Spatial Sciences, Moscow, ID, USA</i> Competitive teaching and research assistantships for graduate students.	
Global Leadership Full-tuition Scholarship	2015-2019
<i>Coe College, Cedar Rapids, IA, USA</i> Competitive full-tuition scholarship awarded to one international student per class.	
DeJong Biology Research Award	2019
<i>Coe College Dept. of Biology, Cedar Rapids, IA, USA</i> Award for outstanding undergraduate research in Biology	
Phi Beta Kappa, Phi Kappa Phi	2019
<i>National Honors Society, Coe College, Cedar Rapids, IA, USA</i> Invitations are based on overall GPA and research portfolio.	

RESEARCH & TEACHING

- Lead Researcher, MS thesis** 2021 – 2023
University of Idaho Dept. of Earth & Spatial Sciences, Moscow, ID, USA
Title: Combining Multispectral and Three-Dimensional Data from Drone Imagery to Detect Forest Insect Damage: An Evaluation of a Novel Approach to Identify the Vertical Structure of Damage in Trees in the Northern Rocky Mountains.
- Research Assistant, NASA CSDA project** 2021 – 2023
Washington State University (WSU), University of Idaho, USDA Forest Service
Title: Using commercial satellite imagery to study insect outbreaks in the US: Outbreak characteristics and evaluation of Landsat-based algorithms. PI: Dr. Arjan Meddens, WSU.
- Collaborative Researcher** 2022 – 2023
University of Idaho Dept. of Plant Sciences, Moscow, ID, USA
Title: A comparison and development of methods for estimating sagebrush shrub volume using unmanned aerial systems. PI: Georgia R. Harrison.
- Teaching Assistant** 2021 – 2023
University of Idaho Dept. of Earth & Spatial Sciences, Moscow, ID, USA
Courses: Physical Geography, Intermediate GIS, Hydrological Applications of GIS/Remote Sensing
- GIS Tutor** 2018
Coe College Dept. of Biology, Cedar Rapids, IA
Courses: spatial ecology lab

PROFESSIONAL EXPERIENCE

- Remote Sensing and GIS Specialist** 2024 – 2025
RedCastle Resources, Inc., Salt Lake City, UT, USA
Federal contractor to the USDA Forest Service's Geospatial Office (GO). Responsibilities: conduct remote sensing and geospatial analyses for resource mapping and forest health assessments at various spatial and temporal scales. Utilize programming (R, Python, JavaScript), Git/GitHub, GIS tools, and cloud platforms to develop operational workflows.
- Quality Control Lab Analyst** 2019 – 2021
Archer Daniels Midland (ADM), Cedar Rapids, IA
Responsibilities: perform daily analytical and blind sample tests, perform maintenance, calibration, and troubleshooting of all analytical equipment, review and validate standard operating procedures.
- Undergraduate Lab Assistant** 2016 – 2019
Coe College Dept. of Biology, Cedar Rapids, IA
Responsibilities: prepare samples and solutions for all biology lab courses, maintain and update equipment for field courses, maintain inventory of chemical stockroom.
- IT Technician** 2018
Coe College Office of Information Technology, Cedar Rapids, IA
Responsibilities: update and maintain hardware and software of all computers, printers, provide technical support (in-person and phone)

SKILLS & PROFICIENCIES

Programming Languages

R, Python, JavaScript, HTML/CSS, SQL

Software & Tools

ArcGIS Pro, QGIS, ERDAS Imagine, CloudCompare, Agisoft Metashape, Pix4D, Open Drone Map, Google Earth Engine, Google Collab, Google Cloud Platform, RShiny, RMarkdown, Jupyter Notebooks, ArcPy Notebooks, Git, GitHub, VSCode, RStudio, Emlid Studio, Windows batch scripting, Bash scripting

Languages

English, Nepali, Hindi (written & spoken), Urdu (spoken)

Certificates

Geospatial Information Systems (GIS) Graduate Certificate (Dept. of Earth & Spatial Sciences, University of Idaho, Moscow, ID, USA)

REFERENCES

The references are grouped according to career and academic stages. The relation of the listed reference to the applicant is in parentheses.

Academic references

Dr. Jeffrey Hicke (Graduate advisor, MS thesis committee member, collaborative researcher), Professor, Department of Earth and Spatial Sciences, University of Idaho, Moscow, ID. Email: jhicke@uidaho.edu. Phone: (+1) 208-885-6240

Dr. Jason Karl (MS thesis committee member, collaborative researcher), Associate Professor of Rangeland Ecology and Harold F. and Ruth M. Heady Endowed Chair of Rangeland Ecology, Department of Forestry, Rangeland, and Fire Sciences, University of Idaho, Moscow, ID. Email: jkarl@uidaho.edu. Phone: (+1) 208-885-0255

Dr. Arjan Meddens (MS thesis committee member, collaborative researcher), Assistant Professor, School of the Environment, Washington State University, Pullman, WA. Email: arjan.meddens@wsu.edu. Phone: (+1) 509-335-8570

Dr. Karen Humes (Graduate teaching assistant supervisor), Professor, Department of Earth and Spatial Sciences, University of Idaho, Moscow, ID. Email: khumes@uidaho.edu. Phone: (+1) 208-596-5725

Professional references

Lila Leatherman (Peer Mentor), Technical Team Lead, RedCastle Resources, Inc., Salt Lake City, UT. Email: lila.leatherman@gmail.com

Anna Peterson (Supervisor and Project Manager), Geospatial Project Manager, RedCastle Resources, Inc. Salt Lake City, UT. Email: apeterson5433@gmail.com

Eric Rounds (Project Manager), Remote Sensing Project Manager, RedCastle Resources, Inc., Salt Lake City, UT. Email: eric.rounds@redcastleresources.com